

Network Monitoring and Threat Detection Lab

Objective

The purpose of this lab was to use Wireshark to capture and analyze network traffic. The goal was to identify normal traffic and detect suspicious or attack-like behavior such as ping sweeps, port scans, and high volume ICMP traffic.

Traffic Capture Setup

Wireshark was used to capture live network traffic on the active network interface. Normal traffic was first generated using basic commands to establish a baseline for comparison.

ICMP Ping Sweep

A ping sweep was performed by sending ICMP echo requests to multiple IP addresses on the local network. This simulates network scanning behavior.

In Wireshark, this was identified by:

- A single source IP sending requests
- Multiple destination IP addresses

- ICMP protocol traffic

No.	icmp icmpv6	Source	Destination	Protocol	Length	Info
3954	167.519674	192.168.0.1	192.168.0.25	ICMP	74	Echo (p
3955	167.575533	192.168.0.25	192.168.0.2	ICMP	74	Echo (p
3956	167.630840	192.168.0.2	192.168.0.25	ICMP	74	Echo (p
3957	167.703942	192.168.0.25	192.168.0.3	ICMP	74	Echo (p
3958	167.708297	192.168.0.3	192.168.0.25	ICMP	74	Echo (p
3959	167.770975	192.168.0.25	192.168.0.4	ICMP	74	Echo (p
3960	167.774072	192.168.0.4	192.168.0.25	ICMP	74	Echo (p
4035	173.546275	192.168.0.25	192.168.0.9	ICMP	74	Echo (p
4099	179.045893	192.168.0.25	192.168.0.11	ICMP	74	Echo (p

Frame 280: Packet, 74 bytes on wire (592 bytes captured)	0000	2c 99 24 90 e5 d0 f4 b3 01 8e 93 74
Ethernet II, Src: Intel_8e:93:74 (f4:b3:00:00:00:00), Dst: Intel_8e:93:74 (f4:b3:00:00:00:00)	0010	00 3c a2 fd 00 00 80 01 00 00 c0 a8
Internet Protocol Version 4, Src: 192.168.0.25, Dst: 192.168.0.2	0020	00 01 08 00 4c b6 00 01 00 a5 61 62
Internet Control Message Protocol	0030	67 68 69 6a 6b 6c 6d 6e 6f 70 71 72
	0040	77 61 62 63 64 65 66 67 68 69

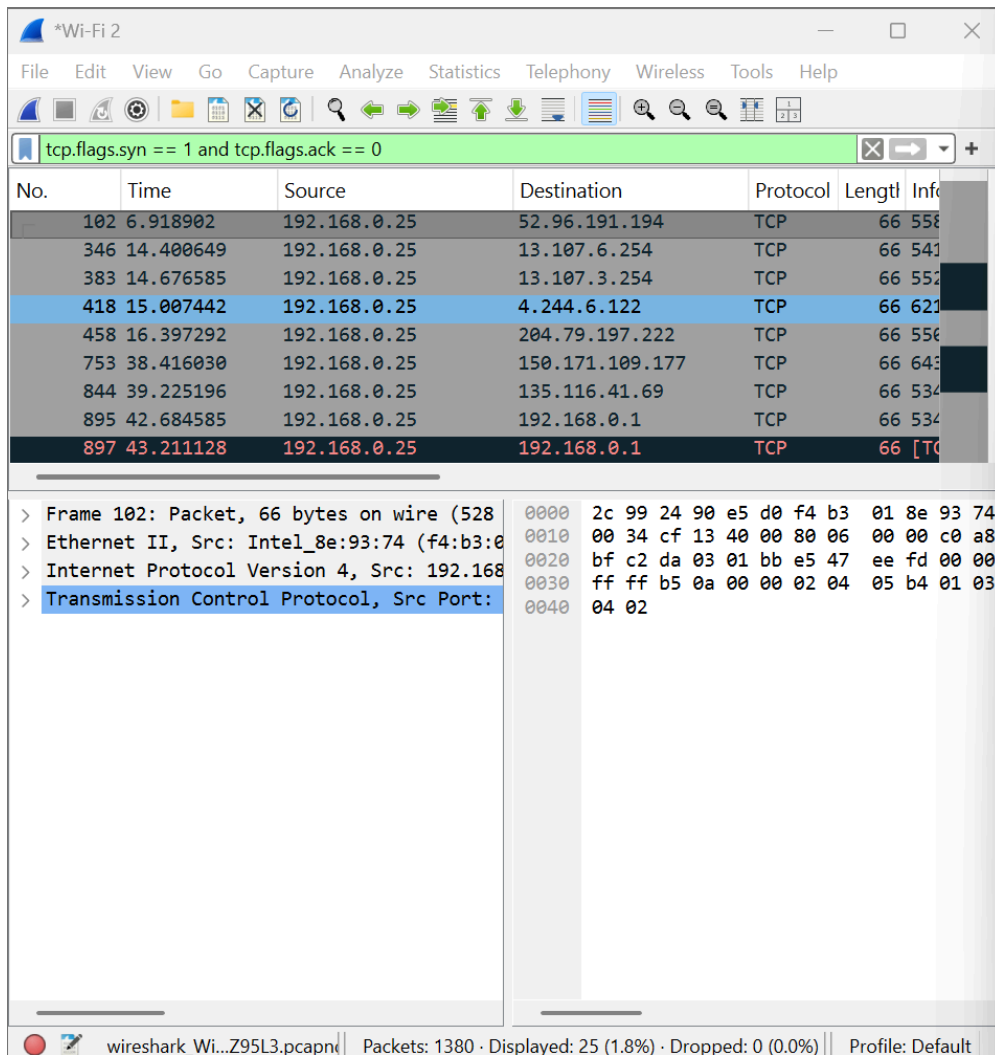
TCP SYN Scan

A port scan was simulated using PowerShell to send TCP SYN packets to multiple ports on a target device.

In Wireshark, this was identified by:

- Multiple SYN packets
- Same destination IP

- Different destination ports



ICMP Flood (DoS-like Activity)

A high-rate ICMP flood was generated using large packet sizes. This simulates denial-of-service behavior.

In Wireshark, this was identified by:

- A large number of ICMP packets
- Repeated communication between the same devices
- Larger packet sizes than normal

Capturing from Wi-Fi 2

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	icmp icmpv6	Source	Destination	Protocol	Length	Info
67	2.057058	192.168.0.1	192.168.0.25	ICMP	124	Destination Unreachable
141	4.013930	192.168.0.1	192.168.0.25	ICMP	124	Destination Unreachable
162	5.517652	192.168.0.1	192.168.0.25	ICMP	124	Destination Unreachable
322	7.016378	192.168.0.1	192.168.0.25	ICMP	124	Destination Unreachable
504	11.040565	192.168.0.25	192.168.0.1	ICMP	74	Echo (ping) request
505	11.046492	192.168.0.1	192.168.0.25	ICMP	74	Echo (ping) reply
553	11.507067	192.168.0.1	192.168.0.25	ICMP	124	Destination Unreachable
638	13.010641	192.168.0.1	192.168.0.25	ICMP	124	Destination Unreachable

> Frame 26: Packet, 124 bytes on wire (992 bytes captured) on interface 0

> Ethernet II, Src: Commscope_90:e5:d0 (2c:90:e5:d0:90:e5:d0), Dst: 08:00:00:00:00:00

> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0

> Internet Protocol Version 4, Src: 192.168.0.25, Dst: 192.168.0.1

> Internet Control Message Protocol

```

0000 f4 b3 01 8e 93 74 2c 99 24 90 e5 d0
0010 08 00 45 c0 00 6a c0 ea 00 00 40 01
0020 00 01 c0 a8 00 19 03 03 7e b3 00 00
0030 00 4e a3 ca 00 00 80 11 15 6a c0 a8
0040 00 01 00 89 00 89 00 3a 9f bd 9f 06
0050 00 00 00 00 00 00 20 43 4b 41 41 41
0060 41 41 41 41 41 41 41 41 41 41 41 41
0070 41 41 41 41 41 41 41 00 00 21 00 01
  
```

Internet Control Message Protocol: Protocol | Packets: 2413 · Displayed: 57 (2.4%) | Profile: Default

Conclusion

In this lab, Wireshark was successfully used to capture and analyze both normal and suspicious network traffic. Different types of attack-like behaviors were identified, including ping sweeps, port scans, and ICMP floods. This demonstrates how network monitoring tools can be used to detect potential security threats in real time.